

NICHOLAS SOBCHAK

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[LinkedIn](#) | [GitHub](#) | U.S. Citizen, Eligible for U.S. Security Clearance

EXPERIENCE

Founding Software Developer – ZipTip

Dec 2025 – Present

Remote / Westminster, MD

- > Developed and implemented authentication and authorization services for ZipTip and Stripe accounts using Cognito.
- > Integrated RESTful APIs for seamless communication between frontend and backend services using boto3.
- > Built and ran automated tests for backend services using pytest.
- > Used AWS SAM to locally test and validate Lambda functions before deployment.

SKILLS

C++: Crow, Catch2, SQLite, JSONL

Python: AWS (Lambda, SAM CLI), boto3, Cognito, pytest, requests, JSON

Databases: SQL, DynamoDB

Tools: GitHub, Linux, Vim, Tmux, CMake, Docker, Valgrind

TECHNICAL PROJECTS [GitHub](#)

QuickQuill – C++17, Crow, SQLite3, Catch2

Dec 2025 – Present

- > Built a 1.4 million word dictionary API with a Trie-based backend in C++, supporting a **sub-XXms** autocomplete feature. Developed a RESTful backend using Crow web framework to serve web-based clients (**[LINK HERE](#)**).
- > Utilized Python to populate the SQL database, and designed a database class for real-time data management and optimized database queries.
- > Implemented in-memory caching, reducing repeated query latency from > 200ms to < 0.18ms (~1350x speedup) for popular queries.
- > Engineered a memory-safe Trie backend with smart pointers. Constructed automated unit and end-to-end tests in a CI/CD pipeline using GitHub Actions.

LEADERSHIP AND INTERESTS

Ice Hockey > Team Captain; Two-time State Champion; Two-time National Qualifier

Orchestra > Section Leader; demonstrated collaboration, discipline, and adaptability

EDUCATION

University of Maryland, Baltimore County

Aug 2025 – May 2027

Bachelor of Science in Computer Science (Expected)

Carroll Community College

Sept 2023 – May 2025

Associate of Arts in Computer Science, GPA: 3.6 Overall, 4.0 Major